YEAR 7 · UNIT 4 REVIEW · SCIENCE UNIT 3 · HOMEWORK 10

# Faster in January

Part 1: Maths — Unit 4 review
Part 2: Science — 3.2, 3.3 and the football model, with decimals
Part 3: Reading graphs with decimals

**Name:** _____ **Class:** _____

**Date given:** _____ **Date due:** Friday 30 October 2026

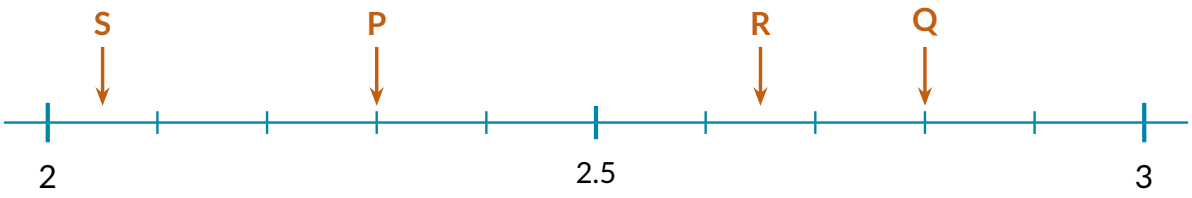
**How to do this homework**

- **Sections A–C:** maths (Unit 4). **D–F:** science, and F uses decimals. **G:** reading graphs.
- No calculator. Estimate first. Then write the calculation, then the answer.
- Answer in full sentences when it says so. *"Gravity"* is not a sentence. *"Gravity pulls the planet towards the Sun"* is.

## Section A · Unit 4 Review — Comparing, Adding and Subtracting

### A1 Read the line

The line goes from **2** to **3**. Every small line is **0.1**. **(a)** Write the number each arrow points to.




| | | | |
|---|---|---|---|
| P _____ | Q _____ | R _____ | S _____ |

**(b)** Write the four numbers in order. Start with the **largest**.

_____

### A2 Which is bigger?

Write <, > or = in each box.

*Tip:* compare the whole numbers first. Then add zeros: 0.7 = 0.70, so 0.7 > 0.65.

(a) 0.5 ☐ 0.45

(b) 2.1 ☐ 2.09

(c) 8.08 ☐ 8.8

(d) 6.4 ☐ 6.400

(e) 15.6 ☐ 9.87

(f) 1.25 ☐ 1.3

## A3 Line up the points

**Written method**

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. The point in the answer goes **straight down**.

**(a)** 7.8 + 4.56

**(b)** 23.4 + 0.87 + 5

**(c)** 16.25 + 3.9

**(d)** 10 − 3.42

**(e)** 6.3 − 2.75

**(f)** 0.9 − 0.36

## Section B · Unit 4 Review — Multiplying and Dividing

**Where the point goes**

**Multiply** Take the point out and multiply. Then **estimate** to put it back.
*Example:* 3.8 × 12 → 38 × 12 = 456 → estimate 4 × 12 = 48 → **45.6**

**Divide** Short division. The point goes **straight up**. If a digit is too small to divide, write **0**. Dividing by a 2-digit number? Write its **times table** first.

## B1 Point out, point back

Fill in the table. Row 1 is done for you.

| Calculation | Estimate | Multiply without the point | Answer |
|---|---|---|---|
| 2.4 × 13 | 2 × 13 = 26 | 24 × 13 = 312 | 31.2 |
| 0.6 × 9 | | | |
| 3.45 × 6 | | | |
| 7.2 × 15 | | | |
| 2.8 × 23 | | | |

## B2 The point goes straight up

Use short division in each box. For **(e)** and **(f)**, write the 15 and the 24 times tables first.

**(a)** 9.45 ÷ 5

**(b)** 4.32 ÷ 4

**(c)** 0.72 ÷ 9

**(d)** 7.3 ÷ 2

**(e)** 52.5 ÷ 15

**(f)** 81.6 ÷ 24

## Section C · Unit 4 Review — Words, Quick Ways and Money

## C1 From English to a calculation

**Word help — the words that choose the operation**

| | |
|---|---|
| **each** | for one of them. *3 pens at 2 dollars each*: 3 × 2 |
| **altogether** | all of it, the total |
| **twice as far** | 2 times as far |
| **What must you add to 4 to make 7?** | 7 − 4. It says *add*, but you **subtract** |
| **3 less than 10** | 10 − 3. The **second** number comes **first** |

Write the **calculation first**. Then write the answer, with its unit if it has one (m, km).

(a) What must you add to 4.65 to make 7?

**Calculation:** _____ **Answer:** _____

(b) A ribbon 2.4 m long is cut into 16 equal pieces. How long is each piece?

**Calculation:** _____ **Answer:** _____

(c) Mr Bowen walks 2.5 km on Monday. On Tuesday he walks twice as far. How far does he walk altogether?

**Calculation:** _____ **Answer:** _____

(d) What is 3.5 less than 10?

**Calculation:** _____ **Answer:** _____

## C2 Now you write the question

Write a word problem for $12 \times 2.40 = 28.80$

**Your word problem must have: tick each one**

- ☐ a real situation
- ☐ a word from C1
- ☐ full sentences
- ☐ units (dollars, kg)
- ☐ a question at the end
- ☐ the answer 28.80

## C3 Use a quick way

Use the **same quick way** as the example. Write the quick way, then the answer.

| Quick way | Example | Your turn |
|---|---|---|
| × 9: × 10, then take away one lot | $3.6 \times 9 = 36 - 3.6 = 32.4$ | (a) $4.7 \times 9 =$ _____ = _____ |
| × 11: × 10, then add one lot | $2.3 \times 11 = 23 + 2.3 = 25.3$ | (b) $6.2 \times 11 =$ _____ = _____ |
| **Split** the number | $4.3 \times 6 = 24 + 1.8 = 25.8$ | (c) $5.3 \times 4 =$ _____ = _____ |
| × 0.25 = a quarter of | $28 \times 0.25 = 28 \div 4 = 7$ | (d) $64 \times 0.25 =$ _____ = _____ |
| × 0.07 = ÷ 100, then × 7 | $200 \times 0.07 = 2 \times 7 = 14$ | (e) $300 \times 0.07 =$ _____ = _____ |
| × 0.5 = half of | $7.4 \times 0.5 = 7.4 \div 2 = 3.7$ | (f) $9.6 \times 0.5 =$ _____ = _____ |

## C4 Change from 10 dollars

Money has **2 decimal places**: 7.8 dollars is written 7.80 dollars.

Mr Bowen buys 3 bánh mì at 1.35 dollars each and 2 iced coffees at 1.90 dollars each. He pays with a 10-dollar note. How much change does he get?

**1.** 3 bánh mì: _____ dollars

**2.** 2 iced coffees: _____ dollars

**3.** Total: _____ dollars

**4.** Change: _____ dollars

Working

## C5 Mr Bowen's homework

Only **one** of Mr Bowen's answers is right. Write **right** or **wrong** for each one. If it is wrong, write the correct answer.

| Mr Bowen wrote | Right or wrong? | Correct answer |
|---|---|---|
| (a) 4.5 + 2.7 = 6.12 | | |
| (b) 9.6 ÷ 3 = 3.2 | | |
| (c) 3.40 > 3.4 | | |
| (d) 60 × 0.4 = 2.4 | | |

# Section D · Science 3.2 — How the Solar System Formed

## D1 Key words

Match each word to its meaning. Write the letter (A–F) in the box.

**nebula** ☐
**hypothesis** ☐
**evidence** ☐
**model** ☐
**limitation** ☐
**plane** ☐

**A** an idea that can be tested
**B** one flat surface
**C** a cloud of dust and gas in space
**D** facts from observations or experiments
**E** something a model cannot do, gets wrong or leaves out
**F** a way of showing something that is hard to observe directly

## D2 Support or contradict?

The **hypothesis**: **The Sun and the planets formed from one spinning cloud of dust and gas.**

Does each fact **support** it (agree) or **contradict** it (go against it)? Tick **one** box.

| Fact | supports | contradicts |
|---|---|---|
| All the planets orbit the Sun in the same direction. | | |
| All the orbits are in the same plane. | | |
| Venus spins the opposite way to most planets. | | |
| HL Tauri is a young star with a flat disc of dust round it. | | |
| Uranus spins on its side. | | |
| Most moons orbit their planet in the same direction. | | |

**(b)** Two facts contradict the hypothesis. Is the hypothesis wrong? Answer in a full sentence. Use the words **most of the evidence**.

## D3 From a cloud to the Sun and planets

**(a)** Put the four steps in order. Write **1**, **2**, **3** and **4** in the boxes.

☐ The hot middle becomes the Sun.

☐ Gravity pulls the nebula together. It spins faster.

☐ Dust sticks together and grows into planets.

☐ The nebula flattens into a spinning disc.

**(b)** Why do all the planets orbit the Sun in the same direction? Answer in a full sentence. Start: **"They all formed from …"**

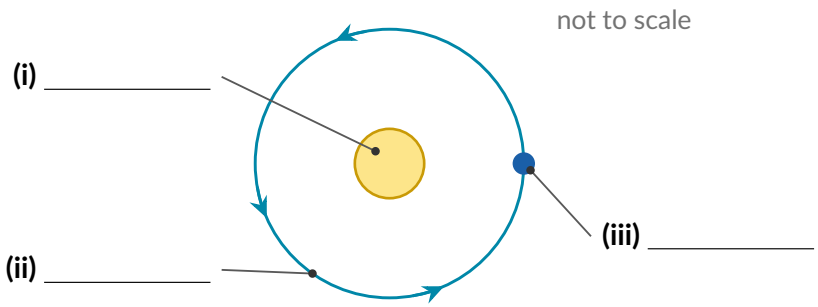
**(c)** Mercury, Venus, the Earth and Mars are small and rocky. Why? Start: **"Near the Sun, it was too hot for …"**

**(d)** The Sun has **99.8%** of the Solar System's mass. What % is left for the rest? _____ %

# Section E · Science 3.3 — Movement in Space

## E1 What keeps a planet in orbit?

**(a)** Label the drawing. Use these words: **Sun** · **orbit** · **planet**




(b) Draw an **arrow** on the planet. Show which way the Sun's **gravity** pulls it.

(c) The arrows show which way the planet moves. Now the Sun's gravity **switches off**. Draw a **dashed line** to show where the planet goes.

## E2 Which forces?

Which forces act on each one? Tick **every** box that is true. Some rows need **two** ticks.

| | gravity | air resistance |
|---|---|---|
| The Earth, orbiting the Sun | | |
| The Moon, orbiting the Earth | | |
| A ball falling from the balcony | | |
| A plane landing on a runway | | |

**(b)** Why do the planets never slow down? Start: **"Space is a …, so …"**

# Section F · Science in Decimals — Faster Near the Sun

**Speed, from Science 3.3**

**Speed**: how fast something moves. **km/s** = kilometres **per** second.
As the distance from the Sun **increases**, the speed of the planets **decreases**. The Sun's pull is **weaker** far away.

## F1 How much faster? How many times as fast?

| | Mercury | Venus | Earth | Mars | Jupiter | Saturn | Uranus | Neptune |
|---|---|---|---|---|---|---|---|---|
| **speed** (km/s) | **48** | **35** | **30** | **24** | **13** | **10** | **7** | **5** |

Fill in the table. Row 1 is done for you. In each row, the **first** planet is the faster one.

| **Planets** | **How much faster?** **subtract**: the difference | **How many times as fast?** **divide**: big ÷ small |
|---|---|---|
| Mercury and Neptune | 48 − 5 = 43 km/s | 48 ÷ 5 = 9.6 |
| Mercury and the Earth | | |
| Mars and Saturn | | |
| The Earth and Mars | | |

## F2 The Sun is a football

In the lesson the Sun became a football, and everything else shrank by the same amount.

| | the Moon | the Earth | Neptune | Jupiter | the Sun |
|---|---|---|---|---|---|
| **Model width** | 0.5 mm | 2 mm | 7.8 mm | 2.3 cm | 22 cm |
| **Like** | a grain of sand | a pinhead | a pea | a grape | a football |

**(a)** Write the five widths in order, **smallest** first. Remember: **1 cm = 10 mm**.

_____

(b) Mr Bowen puts the football against one wall of his classroom, which is **8.5 m** long. The Earth must be **24 m** from the football. How far past the other wall is the Earth?




**Calculation:** _____ **Answer:** _____

(c) Give **one limitation** of the model. Start: **"One limitation is …"**

_____

_____

## F3 Faster in January

The Earth's orbit is *almost* circular. In January the Earth is a little **closer** to the Sun than in July.

| | Distance from the Sun | Speed |
|---|---|---|
| **January** | 147.1 million km | 30.3 km/s |
| **July** | 152.1 million km | 29.3 km/s |

(a) How much closer to the Sun is the Earth in January?

**Calculation:** _____ **Answer:** _____

(b) In which month is the Earth faster? Why? Use the words **closer** and **pull**. Start: **"The Earth is faster in …, because …"**

_____

_____

(c) How much faster is the Earth in January than in July?

**Calculation:** _____ **Answer:** _____

(d) Find the **mean** of the two speeds.

**Calculation:** _____ **Answer:** _____

(e) Round your mean to the nearest whole number. _____

(f) In the lesson, the Earth's speed was **30 km/s**. What do you notice?

_____

_____

## Section G · Reading Graphs with Decimals

### Reading a scale

1. Find two **numbers** next to each other on the scale.
2. Count the **gaps** between them — not the ticks.
3. One tick = the difference ÷ the number of gaps.



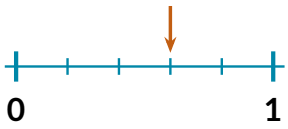

5 gaps: 2 ÷ 5 = 0.4. One tick is **0.4**.

**Word help:** the **axis** is the line along the bottom or side · the **scale** is its numbers · a **tick** is a small line on it · **flat** = not going up or down · **steep** = going up fast

### G1 What is one tick worth?

Write what **one tick** is worth. Then write the number the arrow points to.

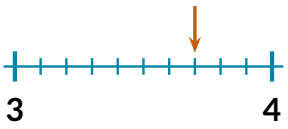
**(a)**




1 tick: _____

Arrow: _____

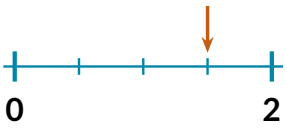
**(b)**




1 tick: _____

Arrow: _____

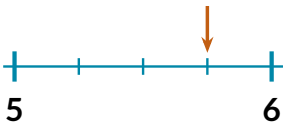
**(c)**




1 tick: _____

Arrow: _____

**(d)**

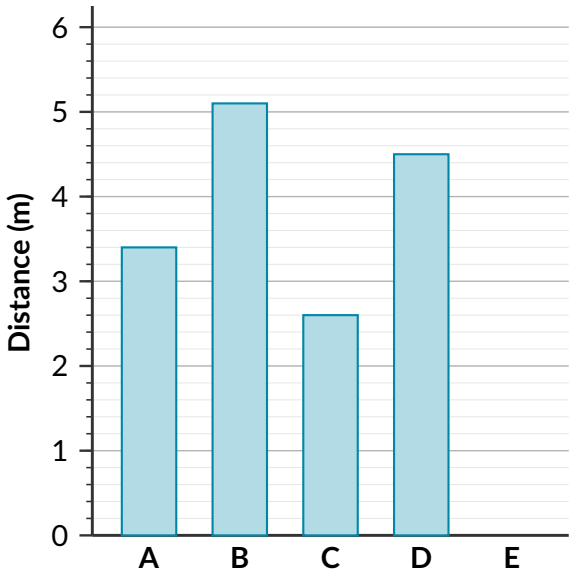



1 tick: _____

Arrow: _____

### G2 Mr Bowen's paper planes

In a long meeting, Mr Bowen made four paper planes. The bar chart shows how far each one flew.




**(a)** One small tick = _____ m

**(b)** How far did each plane fly?

A _____ m B _____ m

C _____ m D _____ m

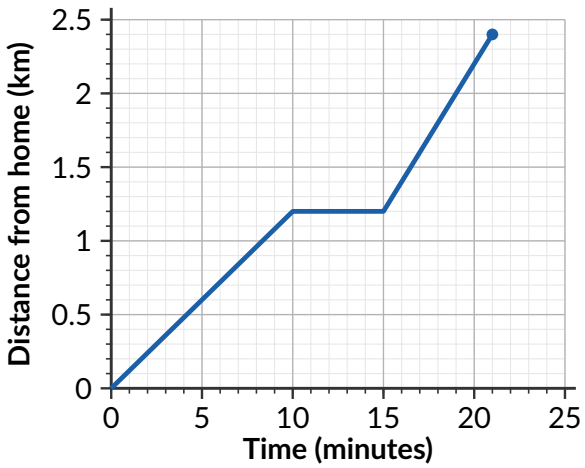
**(c)** How much further did the best plane fly than the worst one?

_____ = _____ m

**(d)** Plane E flew **3.9 m**. Draw its bar.

## G3 Mr Bowen walks to school

Mr Bowen walks from home to school. On the way, he stops at a café. The graph shows how far he is from home.




**(a)** One small tick:
distance _____ km, time _____ min

**(b)** How far is the café from home? _____ km

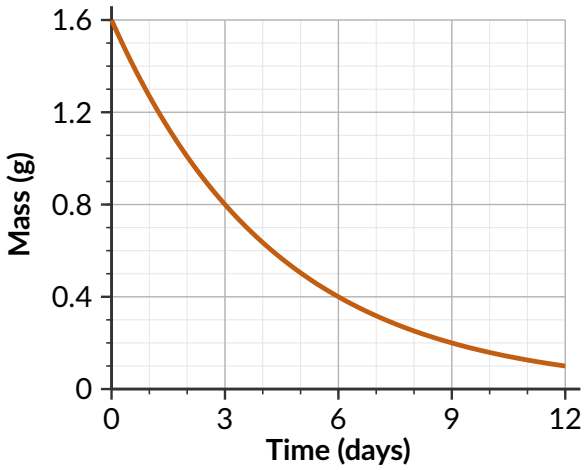
**(c)** How long is he at the café? _____ min

**(d)** How far is school from home? _____ km

**(e)** How far from home is he after **18 minutes**?
_____ km

## G4 A half-life graph

A scientist weighs a sample of a metal that breaks down, like carbon-14 but much faster. The graph shows its mass.




**(a)** One small tick on the mass axis = _____ g

**(b)** The mass at the start: _____ g

**(c)** The mass is 0.8 g after _____ days. So the **half-life** is _____ days.

**(d)** After 6 days: _____ g. After 9: _____ g.

**(e)** After 2 days: about _____ g.

**(f)** Mr Bowen says: *"After 12 days it is all gone."*
After 12 days there is _____ g left.

So he is ☐ right ☐ wrong.

## G5 Tell the story of the graph

Write the story of the **G3** graph in **4 or 5 full sentences**. Tick each one:

☐ three numbers, with units (km, minutes)

☐ what he does at the café, and how long

☐ **faster** before or after the café? How do you know?

☐ a reason he might be faster